

PRIME MOVERS :

Prime mover is an engine or device which converts natural source of energy into mechanical work. In olden days man was depending on his own physical strength and that of animal mainly horse, bullock, therefore, when mechanical systems were invented, the unit of power was horse power.

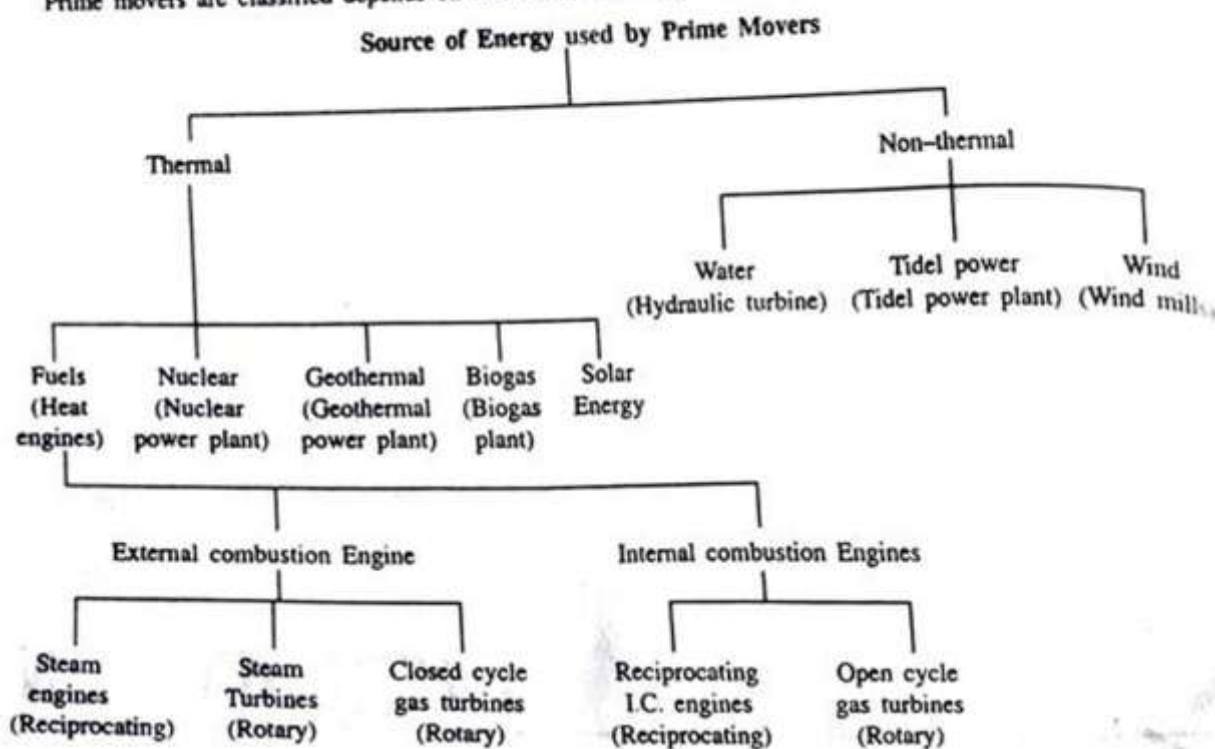
1.2 SOURCES OF ENERGY USED BY PRIME MOVERS :

Prime movers use different sources of energy :

- (a) **Fuels** : When combustion of fuel takes place, heat energy is released depending on its calorific value. This heat energy is converted into mechanical work by a device known as heat engine. It is most widely used source of energy.
- (b) **River Water** : It is another useful form of energy. Water stored at high level possesses potential energy. This potential energy is converted into kinetic energy and ultimately into mechanical work by a prime mover known as hydraulic turbine. River water is also used to generate steam in boilers which in turn used to develop mechanical work using steam turbines and steam engines.
- (c) **Atoms (Nuclear Energy)** : Heat energy is released by fusion and fission reaction of the atoms. This heat energy is used to produce steam which in turn used to produce mechanical work using steam turbines.
- (d) **Solar Energy** : Solar energy is available freely and it is a non polluting source of energy. Solar energy can be used to produce steam to run steam turbine. Using solar photovoltaic cells one can produce electrical energy.
- (e) **Wind Energy** : Kinetic energy of wind is used to produce mechanical work using wind mills. It is also freely available non polluting source of energy.

1.3 CLASSIFICATION OF PRIME MOVERS :

Prime movers are classified depends on the source of energy utilized by them.



What is hydrogen fuel?

Definition:

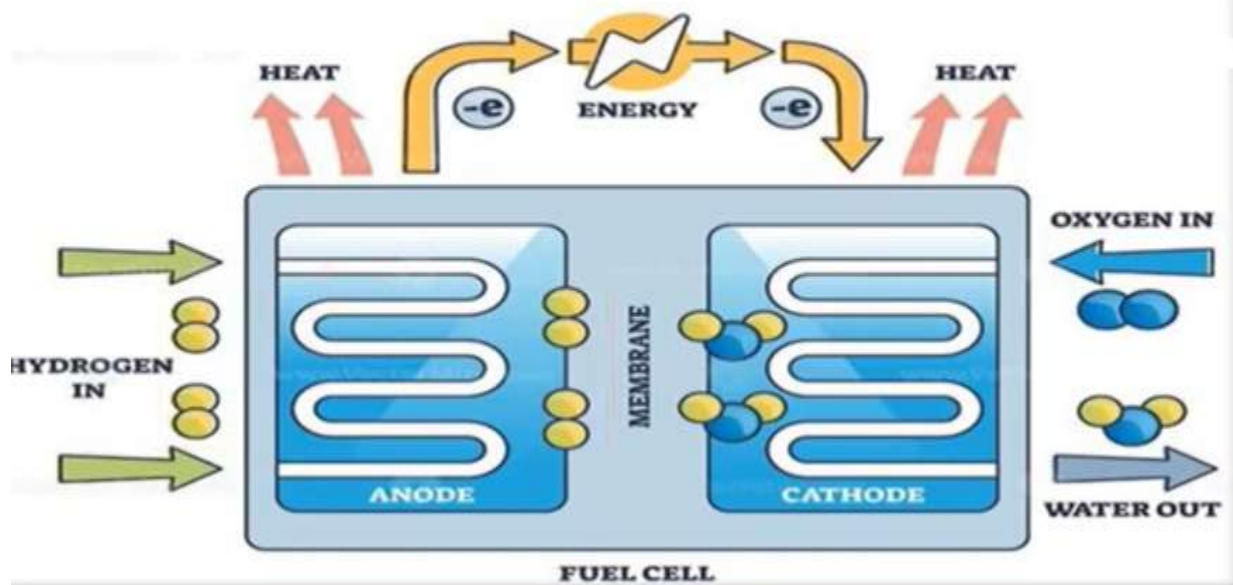
- **Hydrogen fuel is a clean energy source in which hydrogen gas (H_2) is used to produce energy.**
- **When hydrogen reacts with oxygen**
 - ❖ **either by combustion or in a fuel cell**
 - ❖ **it releases energy,**
 - ❖ **and the only by-product is water.**
 - ❖ **So it is an environmentally friendly fuel.**

Working Principle:

➤ Hydrogen fuel can be used in two main ways:

- 1) **Combustion:** Compressed Hydrogen gas is burned in an engine to produce heat energy.
- 2) **Fuel Cell:** Hydrogen reacts electrochemically with oxygen in a fuel cell to produce electricity, heat, and water.

HYDROGEN FUEL CELL



Sources of Hydrogen:

- ❖ Water electrolysis (using electricity)
- ❖ Natural gas reforming
- ❖ Biomass
- ❖ Industrial by-products

Advantages of Hydrogen Fuel:

- 1) Clean fuel (no CO₂ emissions)
- 2) High energy content per unit mass (120 MJ/kg)
- 3) Renewable when produced using green energy
- 4) Suitable for transport, power generation, and space applications

Disadvantages of Hydrogen Fuel:

- 1) High production cost
- 2) Storage and transportation difficulties (requires high-pressure tanks i.e 700 bar)
- 3) Highly flammable
- 4) Infrastructure not well developed (requires a completely new infrastructure for production, distribution by pipelines or by ships and fueling stations.

Application of Hydrogen fuel:

- ❖ Fuel cell vehicles (cars, buses)
- ❖ Power generation
- ❖ Rockets and space missions
- ❖ Backup power systems

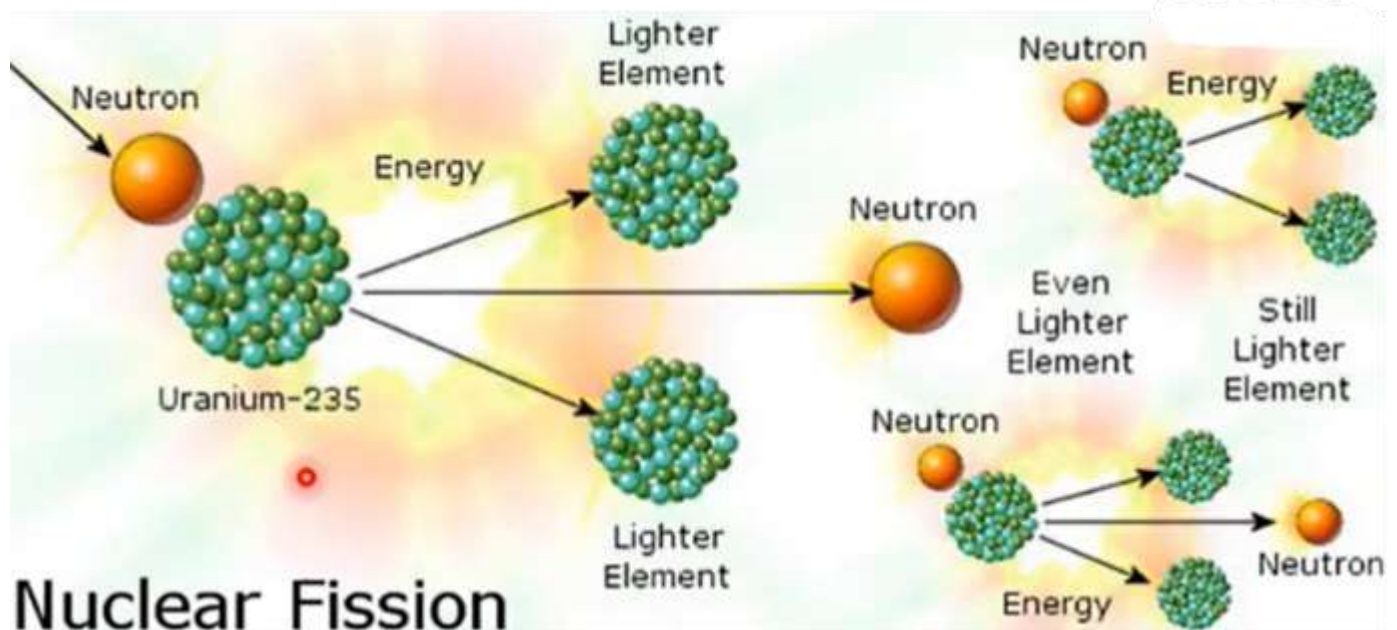
Environmental Impact:

- Hydrogen fuel is environment-friendly, especially green hydrogen, as it helps reduce global warming and air pollution.

Describe nuclear energy.

Working Principle:

- In a nuclear reactor, when a neutron strikes a U-235 nucleus,
 - ❖ it splits into smaller nuclei,
 - ❖ releasing: Heat energy & More neutrons (chain reaction).
- The heat produced converts water into steam,
 - ❖ which rotates a turbine
 - ❖ turbine is connected to a generator which produces electricity.



Advantages of nuclear energy:

- 1) Produces large amounts of energy from small fuel**
- 2) Low greenhouse gas emissions**
- 3) Reliable for base-load power supply**
- 4) Reduces dependence on fossil fuels**

Disadvantages of nuclear energy:

- 1) Radioactive waste disposal problem**
- 2) High initial cost**
- 3) Risk of nuclear accidents**
- 4) Limited availability of nuclear fuel**

Applications of nuclear energy:

- 1) Electric Power Generation**
 - Used in nuclear power plants to generate electricity on a large scale.
- 2) Medical Field**
 - Cancer treatment (radiotherapy)
- 3) Industrial Applications**
 - Non-destructive testing (NDT)
- 4) Agriculture**
 - Food preservation by irradiation (exposing food to controlled doses of ionizing radiation to kill bacteria, insects etc.)
- 5) Space and Defence**
 - Power source for satellites
 - Nuclear submarines and ships

Environmental Impact:

- Nuclear energy causes very low air pollution,
- but improper handling of radioactive waste can cause serious environmental hazards.

Explain difference between renewable and non-renewable energy.

Aspect	Renewable Energy	Non-Renewable Energy
Definition	Energy from sources that are naturally replenished	Energy from sources that are limited and exhaustible
Availability	Inexhaustible	Exhaustible
Examples	Solar, Wind, Hydro, Biomass, Geothermal	Coal, Petroleum, Natural Gas, Nuclear fuels
Environmental Impact	Very low pollution	High pollution and greenhouse gas emission
Global Warming	Helps reduce global warming	Major cause of global warming
Cost (Long term)	Low operating cost	High fuel and maintenance cost
Reliability	Depends on natural conditions	Continuous and reliable
Waste Generation	Negligible	Produces harmful waste
Sustainability	Sustainable (<i>indefinite availability and minimal negative impact</i>)	Not sustainable

Renewable Energy

Definition:

- Renewable energy comes from natural sources that replenish(fill up again) constantly,
 - ❖ like sunlight, wind, flowing water, Earth's heat, and organic matter (biomass),
 - ❖ offering sustainable alternatives to fossil fuels
 - ❖ for electricity, heating, and transportation.



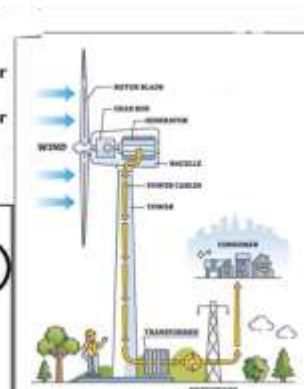
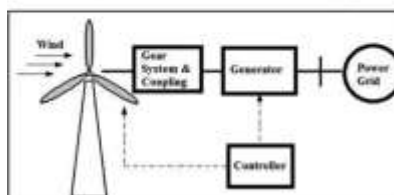
- Renewable Energy can be used repeatedly without depletion.
- It is environmentally friendly and play an important role in sustainable development.

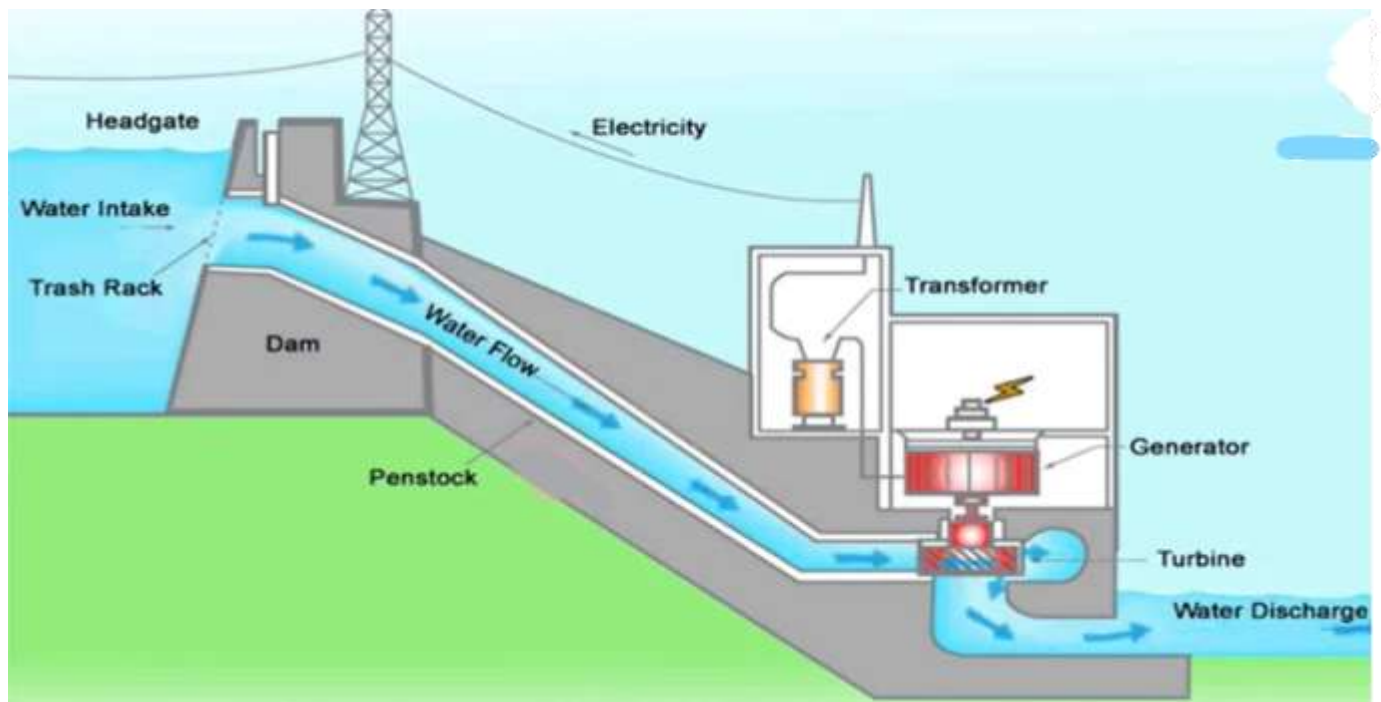


Major Renewable Energy Sources:

2. Wind Energy

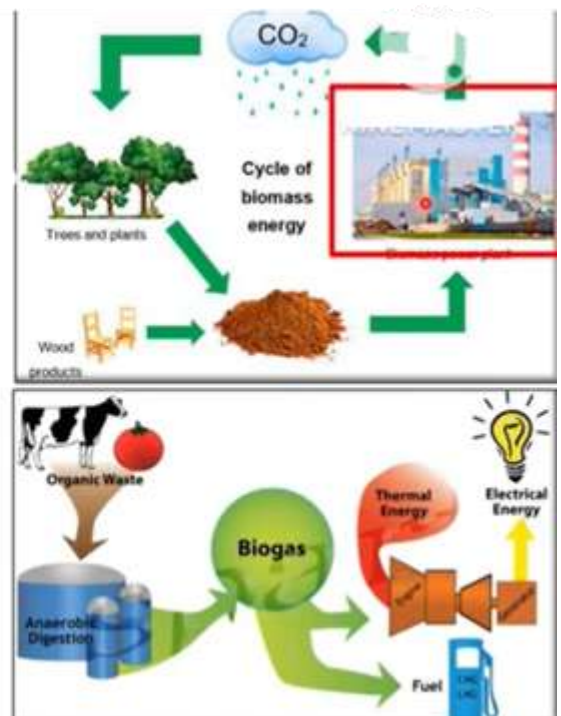
- Energy produced by the movement of air using wind turbines.
- Applications: Power generation, water pumping.





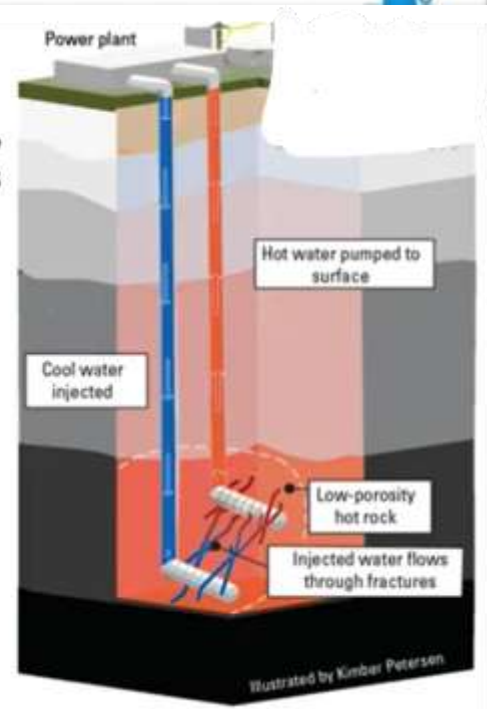
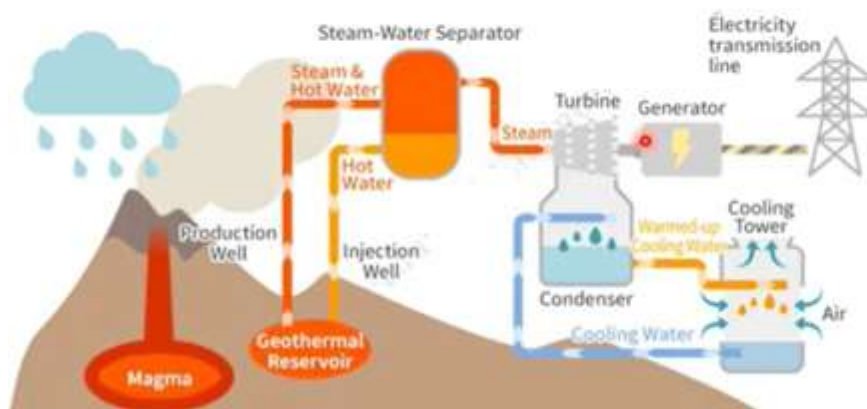
4. Biomass Energy

- Energy derived from organic materials like agricultural waste, wood, and animal dung.
- Biomass, which encompasses organic materials such as agricultural waste, forestry residues, and energy crops, is considered a carbon-neutral energy source.
- When these materials are converted into energy, the carbon dioxide released during the process is offset by the carbon dioxide absorbed by the plants during their growth.
- **Applications:** Biogas plants, biofuels, electricity generation.



5. Geothermal Energy

- Energy obtained from heat inside the Earth.
- The geothermal fluid is taken out of the ground, separated into steam and hot water, and the steam is used to generate electricity.
- **Applications:** Power generation, space heating.



6. Tidal Energy

- Energy produced from ocean tides and waves.
- It is a renewable source that converts kinetic & potential energy from the ocean's regular rise and fall (tides, caused by the Moon/Sun's gravity) into electricity.
- **Applications:** Electricity generation in coastal areas.

Advantages of Renewable energy:

- 1) Clean and eco-friendly
- 2) Reduces greenhouse gas emissions
- 3) Sustainable and inexhaustible
- 4) Reduces dependence on fossil fuels

Disadvantages of Renewable energy:

- 1) High initial cost
- 2) Depends on natural conditions
- 3) Requires large land area (solar, wind)
- 4) Energy storage challenges

Facts of Renewable energy:

- Renewable energy helps in reducing global warming, energy security, and sustainable development.

Advantages of Hydro Power Plant

- 1) **Renewable:** Hydropower is completely renewable, which means it will never run out unless the water stops flowing. In short, Once a dam is constructed, electricity can be produced at a constant rate with low operating cost.
- 2) **Green:** Generating electricity with hydro energy is pollution free. The only pollution occurs during the construction of these massive power plants.
- 3) **Reliable:** Hydroelectricity is very reliable energy. There are very little fluctuations in terms of the electric power. Countries that have large resources of hydropower use hydroelectricity as a base load energy source.

4) Flexible: If electricity is not needed, the sluice gates can be shut, stopping electricity generation. The water can be saved for use another time when electricity demand is high.

5) Safe: Compared to other fossil fuels and nuclear energy, hydroelectricity is much safer. Hydroelectricity produced by plant does not produce green house gases.

6) Tourist attraction: The lake that forms behind the dam can be used for water sports and leisure / pleasure activities. Many times large dams become tourist attractions for visitors.

7) Irrigation: Hydroelectric power plant reservoirs collect rainwater, which can then be used for consumption or for irrigation.

In storing water, they protect the water tables against depletion and reduce damage against floods and droughts.

8) Life span: Life of hydro power plant is 100-125 years. For thermal power plant it is 20-25 years.

9) Efficiency: Efficiency is more & not affected by age.

Disadvantages of Hydro Power Plant

1) Environmental Consequences: The environmental consequences of hydropower are related to interventions in nature due to damming of water, changed water flow and the construction of roads and power lines.

The flooding of large areas of land means that the natural environment is destroyed.

2) Expensive: Building power plants in general is expensive. In fact hydroelectric power plants are more expensive. Dams are extremely expensive to build and must be built to a very high standard.

The high cost of dam construction means that they must operate for many decades to become profitable.

3) Droughts : Electricity generation and energy prices are directly related to how much water is available. A drought could potentially affect this.

4) Limited Reservoirs : There are limited number of suitable reservoirs where hydroelectric power plants can be built and even less places where such projects are profitable.

5) Danger to village people: People living in villages and towns that are in the valley to be flooded, must move out. This means that they lose their farms and businesses.

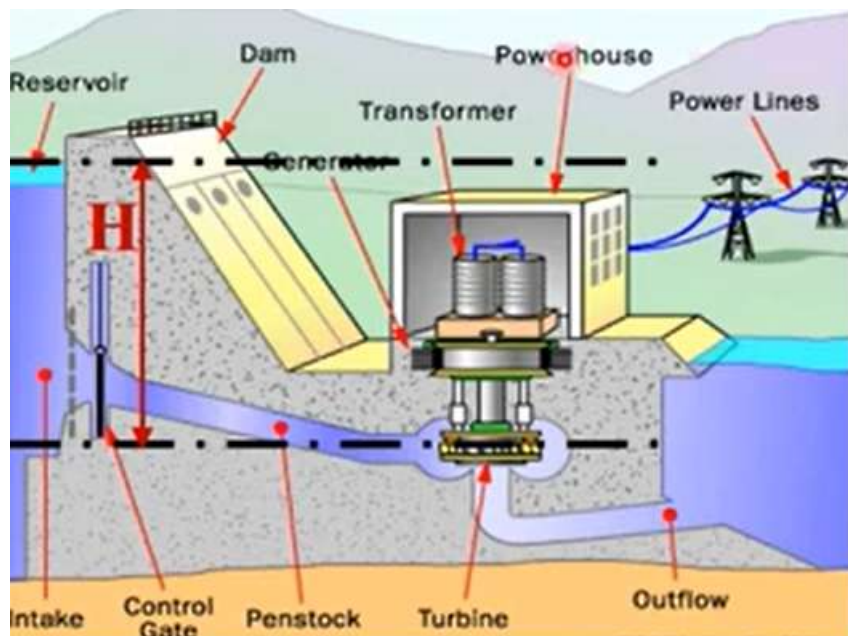
6) Geological damage: The building of large dams can cause serious geological damage. For example, the building of the Hoover Dam in the USA triggered a number of earth quakes.

7) Conflict between countries: Dams built blocking the progress of a river in one country usually means that the water supply from the same river in the following country is out of their control. This can lead to serious problems between neighbouring countries.

Factors for site selection of Hydro Power Plant

1) Availability of water : Main fuel of this plant is water. So, such plant should be located near to river, canal etc. where sufficient water is available all the time. The maximum and minimum quantity of water available throughout the year are also considered.

2) Availability of water head : In order to generate power it is necessary that a large quantity of water at sufficient head should be available. Water head is directly related to the cost of generation of electric power. If effective head is increased, water storage has to be reduced as well as capital cost of the plant is reduced.



3) Water storage capacity : Since there is a wide variation in rainfall during the year, therefore it is necessary to store the water for continuous generation of power. So storage of water in a suitable reservoir or dam has to be placed by a careful geological study of the area.

In short, dam should be located across the river to get continuous water supply throughout the year specially in a dry season.

4) Type of land: The land to be selected for the site should be rocky to withstand the stresses transmitted from the dam structure as well as the thrust of the water when the reservoir is full.

5) Cost of land: Hydro electric power plant needs huge space. So it should be kept in mind that land cost must be cheap.

6) Transportation facilities: Good transportation facilities must be available to any hydro power plant, so that necessary equipment should be reached easily.

7) Distance from the load center : Power plant should be set up near the load centre, this will reduce the cost of maintenance of transmission line. If it is located away from the load center, more transmission line is required to supply the power.

To avoid the line loss and economical power supply, distance of such plant should need more attention.

8) Environment effect: Hydro project submerge large area and many villages. So the environmental effect should also be considered.

What is Ozone Depletion?

Ozone layer depletion is simply the wearing out (reduction) of the amount of ozone in the stratosphere. Ozone depletion means gradual thinning of Earth's ozone layer in the upper atmosphere.

It is caused by the release of chemical compounds containing gaseous chlorine or bromine from industry and other human activities. This thinning is most pronounced in the polar regions, especially over Antarctica.

Importance of ozone layer:

The ozone layer in the stratosphere absorbs a portion of the radiation from the sun, preventing it from reaching the planet's surface.

How it occurs?

Industries manufacture things like insulating foams, solvents, soaps, cooling things like Air Conditioners and Refrigerators use chlorofluorocarbons (CFCs) which are known as **Ozone-Depleting Substances (ODS)**.

These substances are heavier than air, but over time, (2-5years) they are carried high into the stratosphere by wind action.

Ozone depletion begins when CFC's get into the stratosphere. Ultraviolet radiation from the sun breaks up these CFCs. Due to this CFCs releases Chlorine atoms.

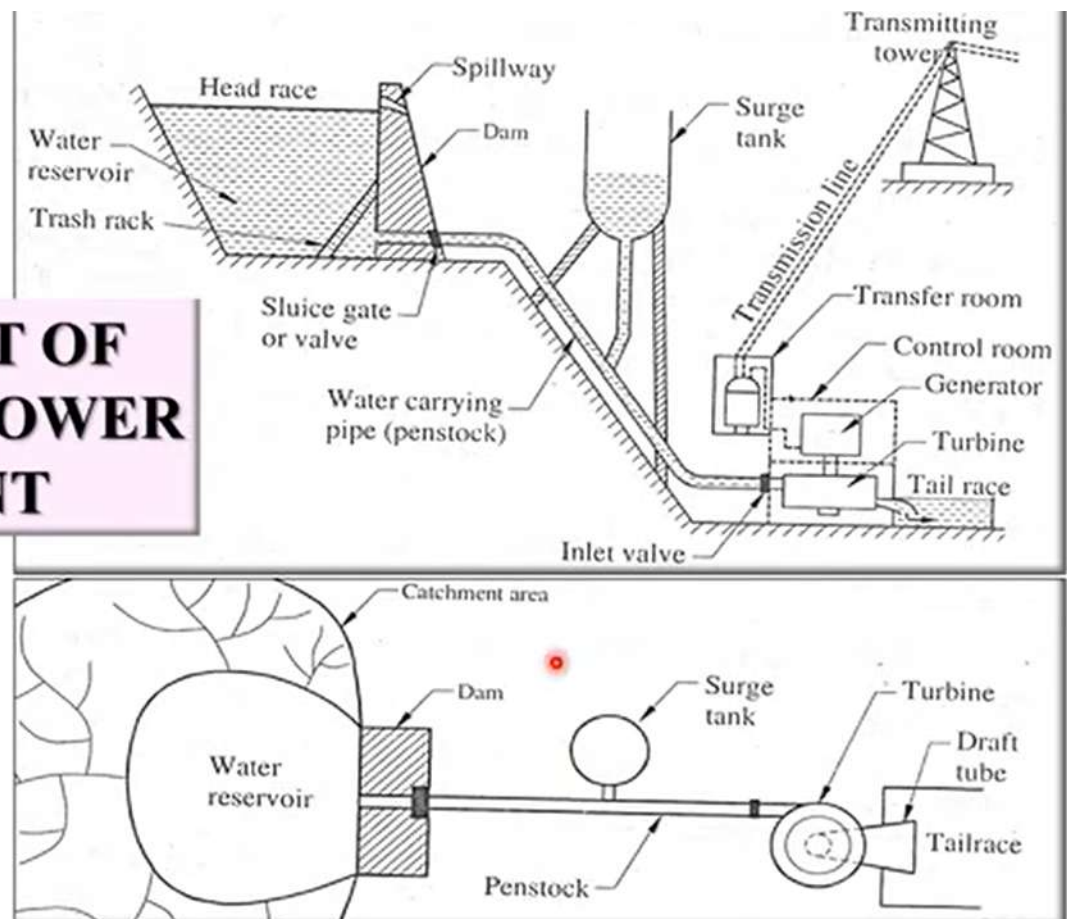
Chlorine atoms react with Ozone, starting a chemical cycle that destroys the good ozone in that area. One chlorine atom can break apart more than 100,000 ozone molecules.

Effects:

Ozone depletion is a major environmental problem because it increases the amount of ultraviolet (UV) radiation that reaches Earth's surface, which increases:

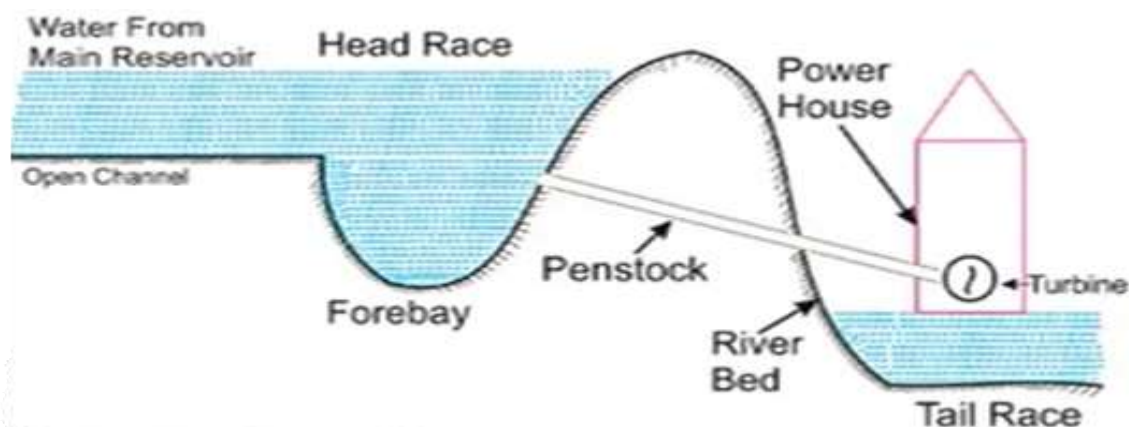
- rate of skin cancer,
- eye cataracts, and
- genetic and immune system damage.

LAYOUT OF HYDRO POWER PLANT



Main Parts of Hydro Power Plant

- 1) **Catchment area:** It is area behind dam across which dam has been constructed.
- 2) **Reservoir:** It is used to store water during rainy season & supply it during dry season.
- 3) **Dam:** It is used to increase height of water level.
- 4) **Trash rack:** It is used to prevent entry of debris which might damage blades & runner of turbine.
- 5) **Surge tank:** It is a small reservoir which provides better regulation of water flow and safe guard against water hammer effect. It is installed along the penstock (between turbine and reservoir). It is also used to serve as a supply tank for increased load conditions, and storage tanks in case of low load conditions.
- 6) **Fore bay:** It is small water reservoir at the end of water passage from the reservoir and before the water is fed to the penstock. It is temporary regulating reservoir. It is used in low head and run off river plant where surge tank is not used.
- 7) **Penstock:** It is used to carry water under the pressure from the large reservoir to turbine.



8) Spillway: To discharge major floods without damage to the dam therefore it is considered as safety valve.

9) Turbine: It is hydraulic machine which converts hydraulic energy (potential/kinetic) into mechanical power by dynamic action of water particles flowing over the blades.

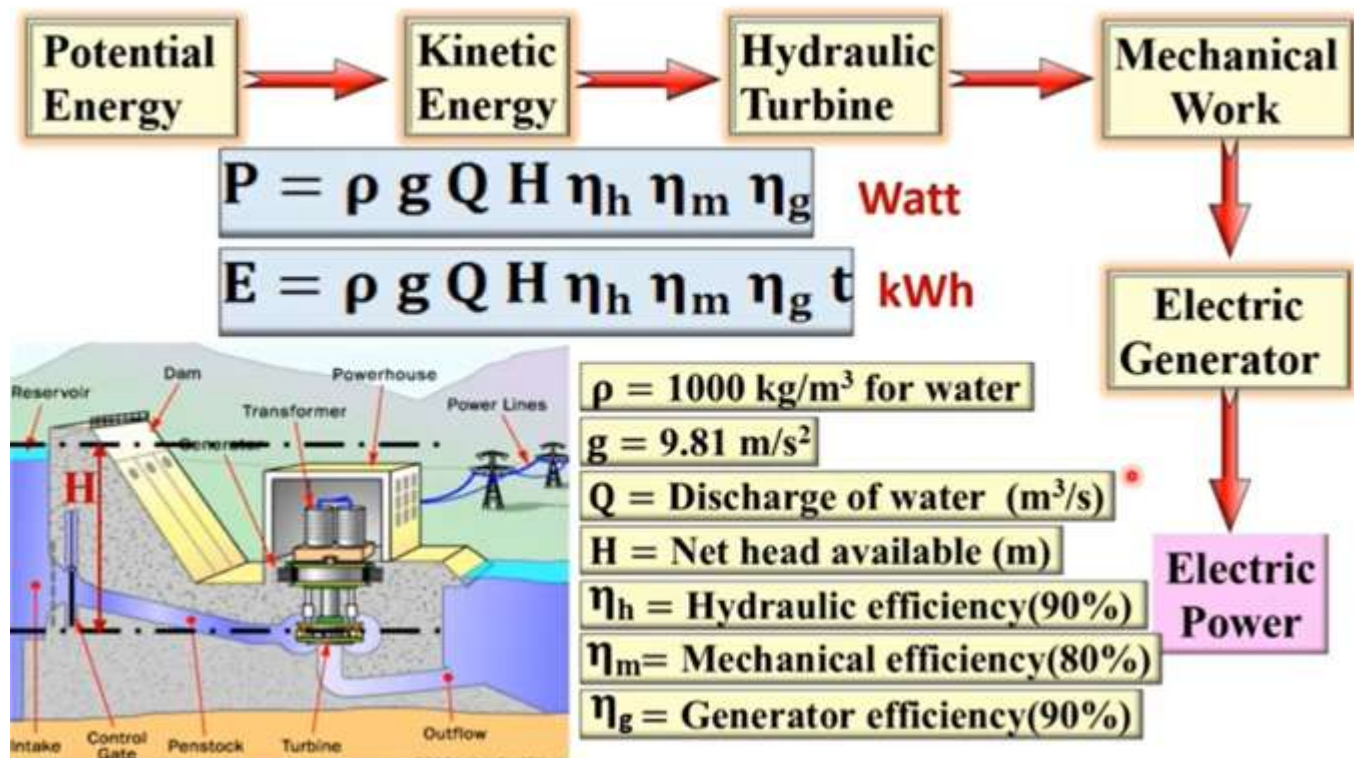
10) Power house: It is building that includes the water turbine, generator, transformer and control room.

11) Draft Tube: It is connected to the outlet of the turbine. It is used to discharge water from exit of turbine to tail race. It is essential part for reaction turbine.

12) Tailrace: It refers to the downstream level of water discharged from turbine.

13) Generator: It is a machine used to convert mechanical energy of turbine into electrical energy.

14) Step up transformer: It converts the Alternating Current (AC) into high voltage current suitable for transmission. It is a transformer that step ups the voltage from the primary winding to the secondary winding while keeping the power same in both the windings at the rated frequency.



Force : Newton's second law of motion defines the force acting on a body. The unit of force is Newton (N). When a force of 1 Newton is applied to 1 kg of mass, it produces acceleration of 1 m/s^2 .

$$1 \text{ N} = 1 \text{ kg m/s}^2$$

The weight (W) of a body is the force with which the body is attracted to the centre of the earth. It is a product of mass of the body and local gravitational acceleration (g).

$$W = mg$$

The value of 'g' at sea level is 9.80663 m/s^2 . For all practical purposes it can be assumed as 9.81 m/s^2 . The weight of a body varies with elevation but the mass remains constant.

Pressure (p) : Pressure is defined as the normal force exerted by a system against unit area of the bounding surface.

In SI system of units, the unit of pressure is Pascal (Pa). One Pascal equals to force of one Newton acting on an area of 1 m^2 .

Pascal is very small unit and very often kilo-Pascal (kPa) and mega-Pascal (MPa) is used.

The other unit of pressure, though not belong to SI system of units are widely being used. They are the 'bar' and the 'standard atmosphere'.

$$1 \text{ bar} = 10^5 \text{ Pa} = 100 \text{ kPa} = 0.1 \text{ MPa}$$

and

$$1 \text{ atm} = 101.325 \text{ kPa} = 1.01325 \text{ bar.}$$

Standard atmosphere is a pressure of atmospheric air at mean sea level. It is defined as the pressure developed by a mercury column of 760 mm. By taking density of mercury equal to 13595.09 kg/m^3 and gravitational acceleration equal to 9.80665 m/s^2 , standard atmosphere will be

$$p = \rho gh = \frac{13595.09 \times 9.80665 \times 0.760}{10^5} = 1.01325 \text{ bar}$$

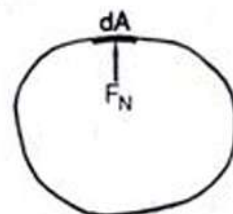


Fig. 1.1 Pressure

Most of the instruments indicate pressure relative to the atmospheric pressure. The pressure relative to the atmosphere is called 'gauge pressure'. The pressure relative to perfect vacuum is called 'absolute pressure'.

Absolute pressure = Gauge pressure + Atmospheric pressure.

When the system pressure is less than atmospheric pressure, the gauge pressure becomes negative. This negative gauge pressure is very often called as vacuum, which is positive number. 20 cm of vacuum means

$$\frac{76 - 20}{76} \times 1.01325 = 0.746 \text{ bar.}$$

Some of the pressure measuring instruments are Bourdon pressure gauge, open U tube manometer, closed U tube manometer (closed limb have vacuum so it measure absolute pressure), Barometer (which measure atmospheric pressure. It is a closed U tube manometer where atmosphere pressure acts on open limb of manometer.)

If H is the difference in the heights of fluid columns in the two limbs of U tube manometer, g is gravitational acceleration and ρ is the density of manometric fluid, then from principle of hydrostatic, the gauge pressure is given by

$$p = \rho g H \left[\frac{\text{kg}}{\text{m}^3} \frac{\text{m}}{\text{s}^2} \text{m} = \text{N} / \text{m}^2 \right]$$

Specific Volume, Density and Specific Gravity : Space occupied by a substance is defined as volume (V). Its unit is m^3 . The specific volume (v) of a substance is defined as the volume per unit mass. Its unit is m^3/kg .

Density (ρ) of a substance is defined as mass per unit volume of the substance. Its unit is kg/m^3 . Thus density is a reciprocal of specific volume.

Specific gravity is defined as ratio of density of a substance at 20°C to density of water at 20°C . Density of water at 20°C is taken as $1000 \text{ kg}/\text{m}^3$.

Energy : Energy is defined as capacity to exert force through a distance. It exists in many forms. Engineering processes involve the conversion of energy from one form to another.

The unit of energy in SI system of units is Nm or Joule (J). The energy per unit mass is the specific energy. Its unit is J/kg.

Power : Rate of energy transfer is known as power. Its unit is Watt (W). It is a small unit and very often kilowatt (kW) and megawatt (MW) is used.

$$1 \text{ W} = 1 \text{ J/s} = 1 \text{ Nm/s}$$

1.5 THERMODYNAMIC SYSTEM AND CONTROL VOLUME :

A thermodynamic 'system' is defined as a quantity of matter or a region in a space upon which attention is focused for the analysis of the problem. Everything external to the system is called 'surrounding' or the environment. System boundary is a real or imaginary boundary which separates the system from surrounding. It is generally denoted by dotted line. System boundary may be fixed or moving.

A system and its surroundings together form a 'universe'. Systems can be classified in three categories (1) closed system (2) open system (3) isolated system. Closed system is one in which there is no mass transfer across the

boundary of the system thus it is a system of fixed mass. Energy interaction can be there across the system boundary. Example of closed system is air trapped in piston-cylinder arrangement as shown in Fig. 1.2. Open system is one in which both mass and energy cross the boundary of the system. Most of the engineering application are open system. Examples of open system are gas turbine, air compressor, pump, nozzle etc. In gas turbine gas at high pressure, high temperature enters in to the turbine and leaves at lower pressure, lower temperature. There are energy transfers across the system boundary.

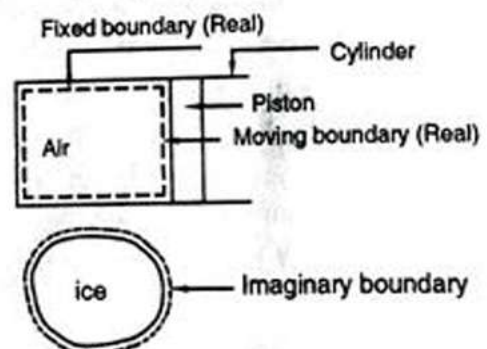


Fig. 1.2 Types of boundary

In thermodynamic analysis of an open system such as gas turbine as shown in Fig. 1.3 attention is focused on certain volume surrounding the gas turbine. This volume is called 'control volume'. The bounded surface is known as 'control surface'. Mass and energy both cross the control surface. Thus there is no difference between open system and a control volume.

An isolated system is one in which neither mass nor energy crosses the boundary of the system.

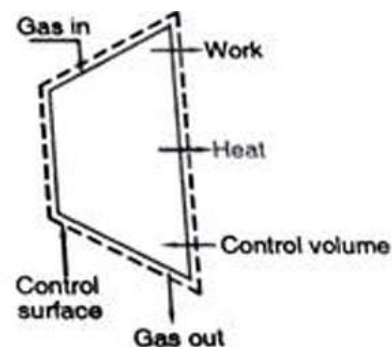


Fig. 1.3 Open system

1.6 PROPERTIES, PROCESS AND CYCLE :

Systems have certain characteristics by which its physical condition can be described, for example pressure, temperature, volume etc. Such characteristics are called 'properties' of the system. Properties are the coordinates to describe the state. There are two types of properties. Intensive properties are independent of the mass of the system. Examples are pressure, temperature, density etc. Extensive properties are dependent of mass. Examples are volume, energy etc. Extensive properties per unit mass, known as specific extensive properties are intensive properties. Example are specific volume, specific energy etc.

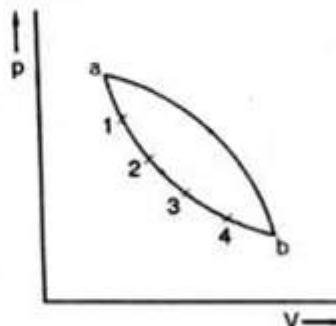
When all the system properties have definite values, the system is said to be in a definite 'state'. In figure 1.4, 'a', '1', etc. are different states of a system.

Any operation in which one or more of the system properties change is called a 'change of state'. In figure 1.4, 'a to 1', '1 to 2' etc. are change of state.

Sequence of states passed through during a change of state is called the 'path' of the change of state.

When path is completely specified, the change of state is known as a 'process'.

A series of change of state such that the final state is identical with initial state is called 'cycle'.



a-1 change of state
a-b path, process
a-b-a cycle

Fig. 1.4 A change of state, a process and a cycle

1.7 TEMPERATURE :

It is a quantitative measure of the degree of coldness or hotness of a system. Its unit in SI system of unit is Kelvin (K). Other units of temperature are Degree centigrade ($^{\circ}\text{C}$) and Degree Fahrenheit ($^{\circ}\text{F}$). The relations between them are as under.

$$\text{K} = ^{\circ}\text{C} + 273.16$$

$$^{\circ}\text{C} = \frac{5}{9} (^{\circ}\text{F} - 32)$$

Temperature is property which distinguishes thermodynamics from other sciences. It helps to distinguish between hot and cold.

1.8 ZEROth LAW OF THERMODYNAMICS :

Zeroth law of thermodynamics is stated as follows.

"When a body A is in thermal equilibrium with a body B, and also with a body C, then B and C must be in thermal equilibrium with each other".

It provides the basis of temperature measurement.